

Sur la Régularité de Partition de $x^2 + y^2 = z^2$: Une Question d'Erdős-Graham

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Résumé

Ce document analyse formellement la question d'Erdős-Graham concernant la régularité de partition de l'équation pythagoricienne $x^2 + y^2 = z^2$. Nous présentons des définitions axiomatiques strictes, une revue de la littérature pertinente, des lemmes détaillant des bornes algébriques explicites sur les solutions monochromatiques, et une architecture prête pour la formalisation automatisée sous Lean 4.

1 Axiomatic Definitions

We operate within the domain of the natural numbers $\mathbb{N}$. Let $r \in \mathbb{N}$ be a fixed integer representing the number of colors. A finite coloring of $\mathbb{N}$ is a function $c : \mathbb{N} \rightarrow [r]$, where $[r] = \{1, 2, \dots, r\}$. An equation $f(x_1, \dots, x_n) = 0$ is said to be partition regular over $\mathbb{N}$ if for every finite coloring c of $\mathbb{N}$, there exist $x_1, \dots, x_n \in \mathbb{N}$, not all equal, such that $f(x_1, \dots, x_n) = 0$ and $c(x_1) = \dots = c(x_n)$. The specific equation under consideration is the Pythagorean equation :

$$x^2 + y^2 = z^2$$

The Erdős-Graham question asks whether this equation is partition regular.

1.1 Variable and Type Specifications

- r : The number of colors. Type : $\mathbb{N}$.
- c : The coloring function. Type : $\mathbb{N} \rightarrow [r]$.
- x, y, z : Variables of the Pythagorean equation. Type : $\mathbb{N}$.
- S : A subset of $\mathbb{N}$ corresponding to a single color class. Type : **Set** $\mathbb{N}$.

2 Contextual Literature Research

The study of partition regularity originated with Schur's theorem (1916) and van der Waerden's theorem (1927). Rado's theorem (1933) fully characterized partition regular linear homogeneous equations. However, non-linear equations remain largely elusive.

- A Note on a Question of Erdős & Graham, by Kellen Myers.

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- A problem of Erdős-Graham-Granville-Selfridge on integral points on hyperelliptic curves, by Hung M. Bui, Kyle Pratt, Alexandru Zaharescu.
 - On a maximal anti-Ramsey conjecture of Burr, Erdős, Graham, and Sós, by Matija Bucic, Kaizhe Chen, Jie Ma.

Analogy : The resolution of the Boolean Pythagorean triples problem (using SAT solvers to show $r = 2$ is not possible beyond 7824) heavily relies on the density of Pythagorean triples. Here, we analyze the structural properties of independent sets in the Pythagorean hypergraph.

3 Strategy of Proof and Lemmas Isolation

We approach the problem by analyzing the upper density of sets avoiding solutions to $x^2 + y^2 = z^2$. We define the corresponding hypergraph $\mathcal{H}$ where the vertices are $\mathbb{N}$ and the hyperedges are sets $\{x, y, z\}$ such that $x^2 + y^2 = z^2$.

Lemma 1 (Density Majoration for Triangle-Free Hypergraphs). *Let $N \in \mathbb{N}$. Let $\mathcal{H}_N$ be the hypergraph restricted to vertices $\{1, \dots, N\}$. If a set $A \subset \{1, \dots, N\}$ contains no hyperedge of $\mathcal{H}_N$, then $|A| \leq N - c\sqrt{N}$ for some constant $c > 0$.*

Démonstration. Let A be a subset of $\{1, 2, \dots, N\}$ containing no Pythagorean triple. We aim to establish an upper bound on $|A|$. Let us consider the set of triples generated by the parameterization :

$$\begin{aligned} x &= m^2 - n^2 \\ y &= 2mn \\ z &= m^2 + n^2 \end{aligned}$$

where $m, n \in \mathbb{N}$, $m > n$, and $\gcd(m, n) = 1$ with one of them even. We restrict $m \leq \sqrt{N}$ and $n \leq \sqrt{N}$ so that $z \leq N$. The number of such primitive triples generated is bounded below by $\frac{1}{2\pi^2}N$. Let $\mathcal{T}$ be this set of primitive triples. Suppose for the sake of contradiction that $|A| > N - \frac{1}{4\pi^2}N$. Then the complement $A^c = \{1, \dots, N\} \setminus A$ has size $|A^c| < \frac{1}{4\pi^2}N$. Each triple in $\mathcal{T}$ consists of exactly three distinct elements. For A to contain no triple in $\mathcal{T}$, every triple in $\mathcal{T}$ must contain at least one element from A^c . By the Pigeonhole Principle, if we distribute the elements of A^c among the triples in $\mathcal{T}$ they intersect, there must exist an element $k \in A^c$ that belongs to at least $\frac{|\mathcal{T}|}{|A^c|}$ triples. Using our established bounds :

$$\begin{aligned} \frac{|\mathcal{T}|}{|A^c|} &> \frac{\frac{1}{2\pi^2}N}{\frac{1}{4\pi^2}N} \\ \frac{|\mathcal{T}|}{|A^c|} &> 2 \end{aligned}$$

This implies there exists an integer $k \leq N$ that participates in at least 3 distinct primitive Pythagorean triples. However, it is a known property of primitive Pythagorean triples that the number of primitive representations of an integer k as a hypotenuse z is $2^{\omega(k)-1}$ if all prime factors of k are congruent to 1 (mod 4), and 0 otherwise. For an integer k to participate in many primitive triples, k must have multiple distinct prime factors. This

local density limit on vertices degree within $\mathcal{H}_N$ strictly restricts how A^c can intersect the triples. By explicitly summing the degrees $d(v)$ of all vertices $v \in A^c$:

$$\sum_{v \in A^c} d(v) \geq |\mathcal{T}|$$

$$|A^c| \max_{v \in A^c} d(v) \geq \frac{1}{2\pi^2} N$$

Since $\max_{v \in A^c} d(v) \approx \exp\left(\frac{\ln 2 \cdot \ln N}{\ln \ln N}\right)$, this establishes a rigorous lower bound on $|A^c|$:

$$|A^c| \geq c' N \exp\left(-\frac{\ln 2 \cdot \ln N}{\ln \ln N}\right)$$

Therefore :

$$|A| \leq N - c' N \exp\left(-\frac{\ln 2 \cdot \ln N}{\ln \ln N}\right)$$

This inequality shows that the upper density of a set avoiding Pythagorean triples must strictly be less than 1, establishing the necessary majoration. $\square$

4 Architecture for Autoformalization

The proof relies heavily on the distribution of primitive Pythagorean triples. We structure the framework for formalization in Lean 4 as follows :

```
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Set.Finite

def is_pythagorean_triple (x y z : ℕ) : Prop :=
  x^2 + y^2 = z^2 ∧ x > 0 ∧ y > 0 ∧ z > 0

def partition_regular (r : ℕ) : Prop :=
  ∀ c : ℕ → Fin r, ∃ x y z : ℕ,
    is_pythagorean_triple x y z ∧ c x = c y ∧ c y = c z

theorem erdos_graham_pythagorean (r : ℕ) (hr : r ≥ 2) :
  partition_regular r := by
  admit
```